

TRAN SI NAM

Computer Vision | Machine Learning | Networks and Telecommunications Intern

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PROFESSIONAL SUMMARY

Senior Electronics and Telecommunications Engineering student with hands-on academic experience in computer vision, deep learning, object detection, dataset curation, model evaluation, and reproducible technical documentation. Built and evaluated a YOLO11 research prototype on a 3,140-image dataset and communicated results through code, model and dataset cards, a thesis, and a defense presentation. Seeking internship opportunities in computer vision, machine learning, computer networks, or telecommunications.

EDUCATION

Bachelor's Program, Electronics and Telecommunications Engineering 2022 to 2026

VNUHCM-University of Science | Faculty of Electronics and Telecommunications | Ho Chi Minh City, Vietnam

- Undergraduate thesis defended on August 15, 2026: *Deep Learning Application for Dangerous Person Detection in Public Areas*.
- Academic focus: computer vision and machine learning, supported by computer networks, data communications, and telecommunications engineering fundamentals.

TECHNICAL SKILLS

Programming and configuration: Python, YAML, Git, GitHub, GitHub Actions

Machine learning and computer vision: PyTorch, Ultralytics YOLO11, deep learning, object detection, model training, inference, image preprocessing, data labeling, data augmentation

Evaluation and experimentation: train-validation-test splits, precision, recall, F1 score, IoU, mAP50, mAP50-95, confusion matrices, error analysis, experiment comparison

Data and compute: Roboflow, Google Colab, NVIDIA Tesla T4, YOLO annotation format

Networks and telecommunications: computer networks, data communications, telecommunications engineering fundamentals

Technical communication: research documentation, model cards, dataset cards, reproducibility guides, responsible-use analysis

PROJECTS

YOLO11 Dangerous-Person Detection in Public Areas

2026

Undergraduate Graduation Thesis | [GitHub repository](#)

Python | PyTorch | Ultralytics YOLO11 | Roboflow | Google Colab | NVIDIA Tesla T4

- Built a two-class object-detection prototype that localizes `normal_person` and `potentially_dangerous_person` instances in public-area images for human-reviewed academic analysis.
- Curated and structured 3,140 images with 10,595 labeled person instances, including 247 self-captured public-area images and hard-negative examples; maintained an 80/10/10 train-validation-test split.
- Trained and compared five YOLO11 variants - YOLO11n, YOLO11s, a YOLO11s preprocessing variant, YOLO11m, and YOLO11l - using a consistent 832-pixel input and held-out evaluation protocol.
- Selected YOLO11s after it achieved 0.87 precision, 0.69 recall, 0.77 F1, 0.80 mAP50, and 0.64 mAP50-95 on 314 held-out test images.
- Packaged configurable training and inference scripts, YAML experiment settings, lightweight tests, checksums, versioned research artifacts, and reproducibility documentation for technical review.
- Documented limitations, potential false positives and false negatives, dataset bias, privacy considerations, and the requirement for human oversight before any operational use.

ADDITIONAL INFORMATION

Languages: Vietnamese (native); English (technical reading and writing)

Availability: Internship-level roles in computer vision, machine learning, computer networks, network communications, and telecommunications